Project: Business Analysis and Modeling for an eBike Rental Startup

Scenario

You will work on a scenario involving a (fictitious) startup company that specializes in renting e-bikes. Mobi-e-rides is a startup that focuses on providing eco-friendly transportation solutions by renting e-bikes to enthusiasts and tourists. The company plans to launch in a metropolitan area with a high demand for sustainable transport options.

Instructions: Use this template to draft the deliverable of the project. You may modify the template to include more information.

Instructions

The tasks in the projects are aligned with the following key generic steps in the process of requirements gathering and modeling:

- Identifying stakeholders for requirement gathering
- Drafting interview questions for gathering requirements
- Employing techniques for requirements prioritization
- Traceability tools for requirements
- Conducting effective interviews
- Facilitating workshops
- Observation techniques
- Designing surveys
- Prototyping
- Use case diagrams
- Mapping out business processes

Step 1: Identifying stakeholders for requirement gathering

Stakeholder	Example	Interest
Customers (enthusiasts)	Few groups of people should be involved: - people who use similar services (for example, renting e-bikes, traditional bicycles, e-scooters, cars) - people who never used similar service and never used e-bikes	Get e-bike in nearest area, rent it easily using simple payment methods (available at least few payment methods and possibility to pay by different currencies), get good quality e-bike (specify with customers) to be able to reach target location (inside the city and some limited area around the city). May conflict with rental agent (for example if e-bike was damaged)

Customers (tourists)	Use two groups as described above	The same as above
Business owners	Startup Project Manager, marketer, accountant, lawyer	Startup Project Manager wants to launch e-bike rental business. He wants to extend variety of options of eco-friendly transport solutions. He wants to deliver service within agreed budget and timelines. May conflict with Investors Marketer wants to promote service to get better profit on business Accountant wants to calculate investments, profits, pay salaries to employees Lawyer wants the rental service met regulation exceptions to provide it legally. May conflict with Local government
Employees (maintenance)	People who fix e-bikes, if they are broken, collect them if they were left not in appropriate place or abandoned	They want to have all e-bikes working and operational on rental station
Employees (rental agent)	People who work at an e-bike rental station, where customers can rent an e-bikes, and get additional equipment (like helmet)	They want to have populated and signed off agreement by customer, that customer is going to use bike in an appropriate way and deliver it back to e-bike rental station in agreed term. May conflict with customer if bike was not delivered, was not delivered in time or with an inadequate level of wear
Local government (for permits and regulations)	Municipal governments, national transport departments, and safety standards bodies	They want that business to be aligned with local law and adherence of government policies.

		May conflict with Startup Project Manager.
Investors	Any people who wish to invest in an e-bike rental business	They want to invest less but make as much profit as possible. They want to invest in eco-friendly solutions. They can conflict with business owners (Startup Project Manager), who take funds from investors and organized e-bike rent business

Step 2: Drafting interview questions for gathering requirements

Questions for customers (Enthusiasts and tourists)	Business owners (Startup PM)	Employees (Rental agents and maintenance staff
What is the most important factor for you when you rent transport? Select relevant options: - price - close availability of rental station - quick availability of e-bike -friendly staff - usage of eco-friendly e-bike - possibility to pay by card - possibility to pay in currency you own (pls specify your currency) -possibility to rent other accessories (for example, helmet, basket, etc.) - other (pls specify)	On which stage are we?	What are the most common situations which may lead to conflicts with customer?
What are your requirements/expectations regarding e-bike? Which characteristics should it meet (for example, lightweight, accumulator, speed, user friendly interface for	What is the target stage?	Which equipment do you need to fix e-bike if it is damaged/broken?

navigation e-bike, eco-friendly, etc.)?		
	What are our resources (human, premises, equipment, documents allowing us to conduct business)?	What are peak hours on rental station?
	Which KPIs are we going to use to identify that business is successful or not on each stage?	

Questions for internal lawyer	Investors
Which regulation must we fulfill to launch business?	What are the most important factors for you when you are making decision to invest?
What is the template of form which customer should fill in before renting e-bike?	Which kind of transport would be preferable for you to extend rental options for customer?
Can we store customer information?	Which conditions/ KPI should we meet to treat that stage may be treated as successful?

Step 3: Employing techniques for requirements prioritization

Must have	Should have	Could have	Won't have
Rental station premises	Equipment to fix e-bikes	Customer loyalty program	Availability of transport usage outside agreed zone/location
Human resources available (lawyer, accountant, maintenance workers, rental agents)	Possibly to scale (open additional rental stations)	Mobile app for rentals	Types of transport which are not eco
All the documents and agreements with Local government	Variety of payment methods	Integration with local tourist attractions	

Paper document to be fulfilled by customer prior renting		Possibility of renting other types of transports	
		Possibility of renting accessories	

Step 4: Traceability tools for requirements

Requirement ID	Description	Stakeholder	Status	Comments
1_M	Rental station premises	Startup PM Lawyer Accountant	In progress	Current stage: Startup PM looking for premises for rental station. Next stages: Layer should create appropriate agreement to rent premises. Accountant should pay the rent
2_M	Human resources available (lawyer, accountant, maintenance workers, rental agents)	Startup PM	In progress	All the staff mentioned should be hired. Currently layer and accountant are hired, but we are still looking for maintenance workers and rental agents
3_M	All the documents and agreements with Local government are ready and in order	Lawyer Local government	In progress	Internal lawyer prepares documents required to start e-bike rental business;
4_M	Paper document to be fulfilled by customer prior renting	Lawyer	Closed	From being fulfilled by customer is created.
5_S	Equipment to fix e-bikes	Maintenance worker	Open	Blocked: no maintenance worker so far
6_S	Variety of payment methods is available	Startup PM Lawyer	Open	Topic to discuss: which payment methods, law compliance

Step 5: Facilitating workshops

Participants: Startup PM, Investor, Lawyer, Accountant **Agenda**

Discuss only Must have features: 1_M – discuss preferable options for rental station premises. 2_M - get an update on onboarding tech staff (maintenance workers and rental agents) 3_M – specify stage of law compliance documents for starting business, identify impediments / blockers Estimate time for completion for every feature Agree time for follow up meeting
Notes **If premises are chosen quickly and some time left, create and discuss low-fidelity floor plan which represents how space can be divided into functional zones (renting area, fixing/ maintenance area, storage, etc.)**

Step 6: Observation techniques

Observations techniques: participant observation.

BA can participate in renting transport service as a customer. Chosen type of transport is bicycle.

Observations outcomes inefficiencies:

- rental station was away from target point (czorsztyńskie jezioro bicycle race)
- some bicycle sizes were not available for rent after 11:00
- it was require to leave some documents (for example, passport). Check if it is legal.
- accessories for safe riding was not available (for example, helmet)

Positive observations outcomes:

-possibility to pay by card

-quick rental process (after signing agreement and choosing bicycle customer is ready to go)

-quick collect process (bicycle was accepted by rental station and documents were returned to customer)

Step 7: Designing surveys

Question 1

What is the most important factor when choosing transport rental service? Chose relevant options:

- **types of transport available**
- **distance to target place**
- **price**
- **possibility to pay by card**
- **other (pls specify)**

Question 2
What kind of transport do you use most for rent?

Question 3
What frustrates you most when you use transport rental services?

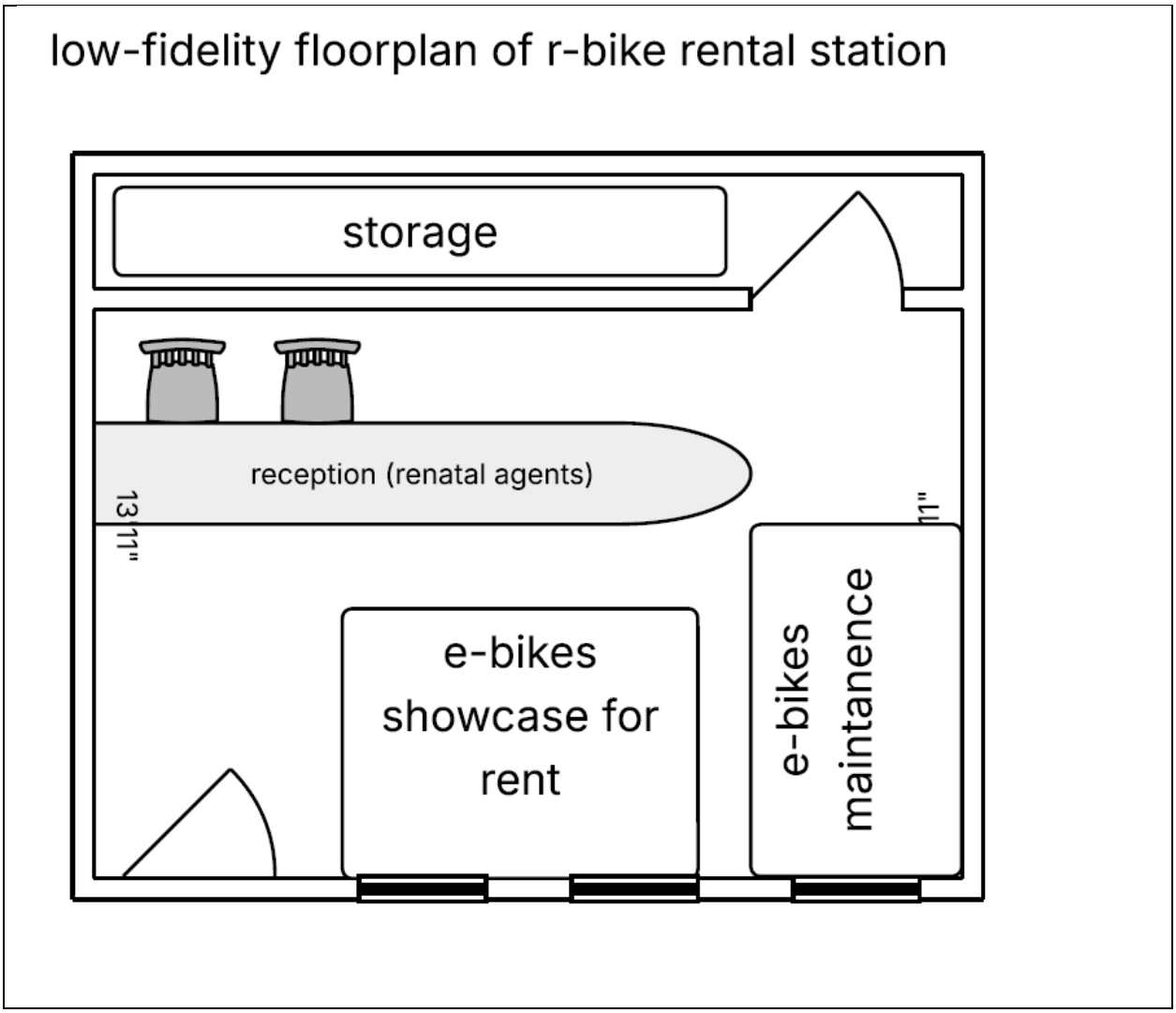
Question 4
What is satisfying when you are renting transport?

Question 5
What is the most preferable way for you to book transport:

- **Physically come to the rental station**
- **By phone**
- **By mail**
- **On-line via website**
- **By mobile app**

Step 8: Prototyping [Optional]

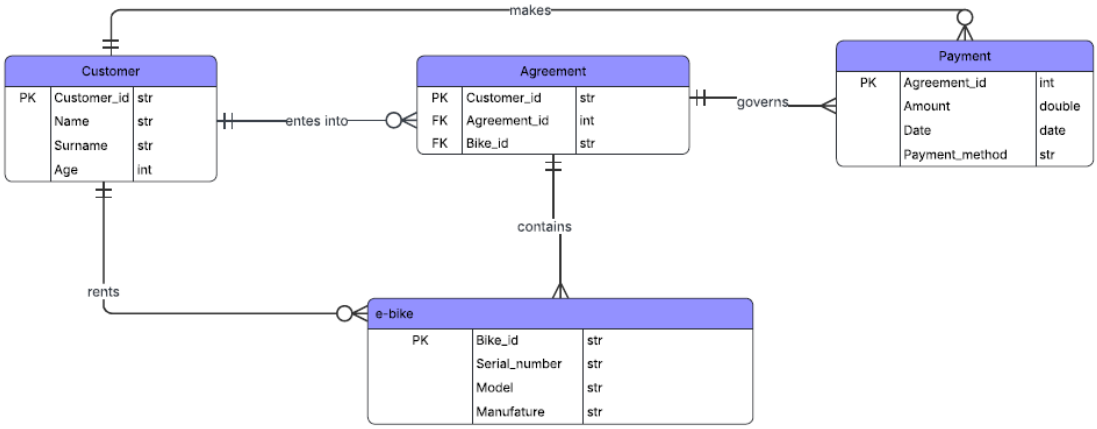
Key features
-showcase
-rent
-maintenance
-storage (equipment and items for maintenance)



Step 9: Modeling business processes

Entity	Attributes
Customer	Customer_id, Name, Surname, Age
Agreement	Customer_id, Agreement_id, Bike_id
Payment	Agreement_id, Amount, Date, Payment method
e-bike	Bike_id, Serial_number, Model, Manufature

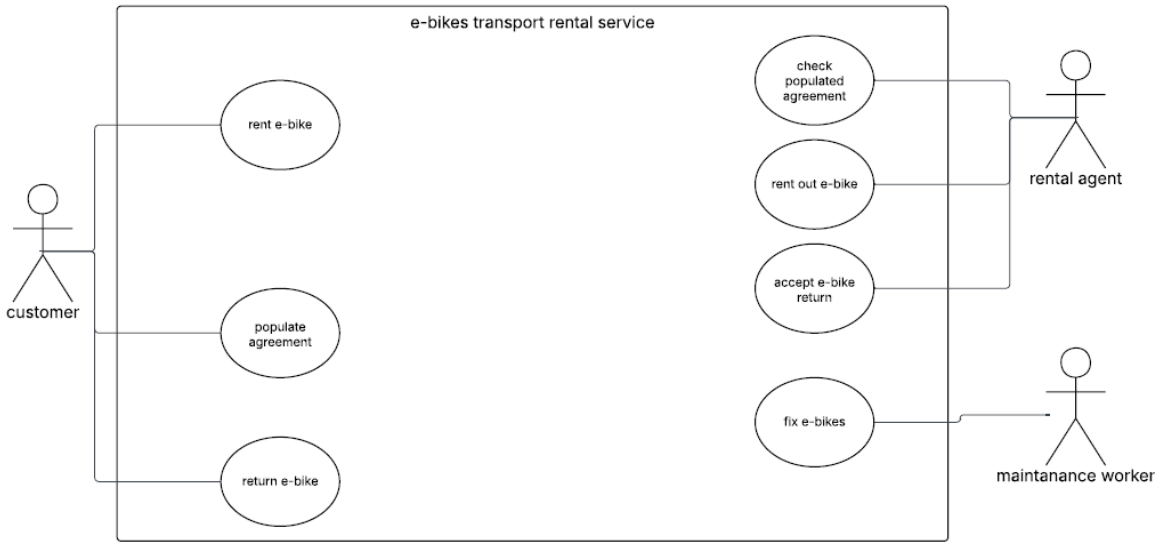
ERD diagram:



Step 10: Use case diagram

Actors: customers, rental agents, maintenance workers, startup PM, lawyer, marketer, accountant, local government, investor
Use cases • rent e-bike • pay by card to rent the bike • return e-bike • populate agreement • check populated agreement • fix e-bikes

Use case diagram:

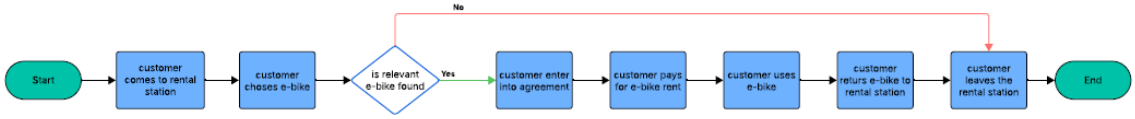


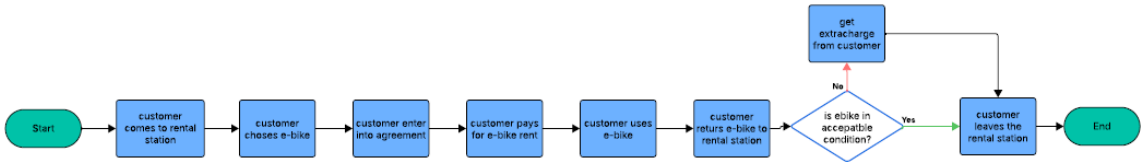
Step 11: Mapping out business processes

Process steps:

- Customer comes to rental station
- Customer chooses e-bike
- Customer enter into agreement
- Customer pays for e-bike rent
- Customer uses e-bike
- Customer returns e-bike
- Customer leaves the rental station

Flowchart:





Step 12: Systems analysis

1. Requirements gathering **Automate user data collection and agreement population**
2. Stakeholder analysis **Hire IT staff who can implement mobile app, create data storage**
3. Current system evaluation **Only physical and manual interventions, automation the process is highly welcomed**
4. Technology assessment **No technologies, manual interventions**
5. Risk analysis **Competitors with more flexible services can take our potential customer**
6. Proposed solutions **Implement mobile app to gather user data and make agreement form population process through the app**

Step 13: Reflection and Validation

The most challenging aspect of this project was its **scope and volume**. Although the estimated duration was four hours, I spent significantly more time completing it. This was mainly because

the provided data served only as a high-level introduction, and many details had to be **assumed or elaborated independently**.

While hints and templates were provided and helped structure the work, the project still required substantial analytical effort. Overall, I consider this a **valuable and practical exercise** for understanding and applying **BABOK concepts** in a realistic scenario.

During this project, I achieved the following:

- I **identified key stakeholders**, described their interests, and considered potential conflicts between them. I deliberately tried to put myself in the *shoes of different stakeholders* to understand their perspectives, expectations, and possible concerns.
- I practiced thinking about **business goals and strategy**, including how the startup could operate, generate profit, and what challenges different stakeholders might face.
- I frequently revisited **course videos and previously completed labs** to refresh terminology, review examples of interview and survey questions, and validate my understanding of concepts.
- I applied the **MoSCoW prioritization technique** for requirements. I found this method intuitive, practical, and easy to use, especially for early-stage product planning.
- The **traceability task** felt somewhat artificial in the lab context. A simple table was sufficient for this exercise; however, in real projects I have used tools such as Jira. In my previous role as a QA, requirements were covered by test cases, but a formal traceability matrix was not explicitly maintained.
- The **workshop facilitation task** reinforced the importance of preparation, time management, and keeping discussions focused on agreed topics and objectives.
- For the **observation task**, I relied on my own experience as a bicycle rental customer, which helped me identify realistic process gaps and improvement opportunities.
- I created a **customer survey**, reusing and adapting several question patterns from earlier labs.
- For **prototyping**, I used Lucidchart. Its functionality is similar to Microsoft Visio, which I am familiar with, particularly from experience in creating technical and spatial diagrams.
- I experienced difficulties with **ERD modeling**, mainly because the course provided only brief textual explanations and lacked clear graphical examples. As a result, ERD notation—especially relationship cardinalities (one-to-one, one-to-many, many-to-many)—was initially unclear to me.

- When creating the **use case diagram**, I initially included only core operational actors (customers, rental agents, maintenance workers). Later, I expanded the diagram to include additional stakeholders. I am still refining my understanding of the appropriate level of detail and stakeholder inclusion for use case diagrams.
- I completed the lab using **Microsoft Word**, which helped me review and correct grammar. However, I am still working on improving my confidence in English, particularly for professional documentation.

Overall, despite the challenges, this project significantly improved my understanding of **business analysis techniques**, modeling approaches, and requirement elicitation practices. I see clear value in applying these skills to real-world projects.